

# Phase-1 Approach — for discussion

My understanding of the deliverable and how I'd approach it, ahead of your technical proposal. This is a discussion input and a phasing view — **not a competing architecture spec**. I'd expect to refine it against your design.

## 1 What I take the deliverable to be

Reading the GO together with our call: the buildable software piece is a **sovereign AI control plane** on top of a managed Kubernetes/OpenShift base — letting defined cohorts (researchers, startups/developers, students) self-serve GPU compute and open-source models, with usage **metered and attributable per tenant**, and the whole thing **operable by the Foundation after handoff**. Hardware/data-center and government-data governance stay out of this lane. Principle: managed-first, portable, open-source, everything-as-code.

## 2 Phase-1 scope

### In scope

- Identity + RBAC (one model, enforced at portal / cluster / API)
- Resource request & approval → GPU/CPU/storage quotas + workspace provisioning
- Model & tool catalog (open-source serving; notebook / Open-WebUI-style access)
- **Usage metering + cost attribution** — the hard core (\$3)
- Audit logging on every action
- **One reference app** — secure RAG over public govt docs (citations; "no source, no answer")

### Out of scope (for now — by design)

- Real government datasets / data-governance (parked until policy is set; kept integration-ready)
- Physical data-center, GPU procurement, NVIDIA MoU, institution setup
- Production security certification & citizen-scale apps
- Fine-tuning for everyone (offer later as a controlled researcher/startup workload)

Personas: go **deep on 2-3** first (researcher, startup/developer, foundation admin); students & departments shown on the roadmap, not built day one.

## 3 Phase-1 core vs. deliberately deferred

The discipline here is the deferral list — these are known and intentionally **not** Phase 1, because each adds attack surface or ops burden before the core is proven.

### PHASE-1 CORE

- Identity + RBAC
- Quota / workspace provisioning
- Model & tool catalog
- Usage metering + cost attribution
- Audit logs
- Guardrails principle (prompt-injection / PII / rate limits)
- Secure-RAG reference app

### DELIBERATELY DEFERRED — PHASE 2+, AFTER A DEDICATED TEAM

- Fine-tuning workspace
- Semantic caching
- Knowledge graph
- Multi-agent orchestration
- MCP / agent-identity (AuthSec-style)
- Hardened LLM firewall
- Govt-data layer

## 4 The hard core — metering & cost attribution

No single tool does this across mixed workloads, so it's the one component I'd treat as first-class: **compute usage** (GPU-hours per namespace, OpenCost/Kubecost-style) + **model serving** (per-token, logged at a gateway in front of the serving stack) → a **unified per-tenant ledger** for showback/billing, with **subsidy rules per cohort** applied there. That ledger is exactly the "at-cost + subsidize chosen cohorts" lever the state wants. Everything else in Phase 1 is integration of open-source parts; this is the real build.

## 5 How I'd de-risk the build

**Not vibe coding — AI-assisted engineering with governance.** Secure-by-design from day one (least-privilege RBAC, audit on every action, input validation, synthetic/public data only — no real AP data in any external tool); maintainability as a first-class deliverable (typed code, tests, ADRs, auto-generated repo docs, GitOps, runbooks) so the Foundation's team can operate it after we step away. AI accelerates scaffolding, tests, and docs; humans own architecture, security, and review. Prototype is explicitly labelled reference-grade — production needs infra/security sign-off, load testing, and staged rollout.

## 6 Open decisions I'd want to align on

- 1 Cloud/vendor + does it expose the **GPU operator & cluster-admin** we'd need (NxtGen/OpenShift vs. others)?
- 2 Source of **identity** — is there a state SSO to federate, or do we own it?
- 3 Who owns **pricing / subsidy policy** (we build the engine; someone sets the rates)?
- 4 **Team shape & cadence** — confirming 1 lead + 3–5 engineers, and the procurement path for them.
- 5 Phase-1 **scale / SLA** target (concurrent users, uptime) — so we right-size, not over-build.

## 7 Role framing & next step

Subject to aligning on role, team, and scope, I'd be glad to help define and lead the **product/platform layer** for Phase 0/1 — scope, personas, the control-plane design, the reference app, and engineer enablement — staying hands-on, with you leading architecture and a dedicated cloud/DevOps engineer owning production operations. Suggested next step: I review your proposal when it's ready and we align on the Phase-1 core + the deferral line on a short call.

AP AI Living Labs · Phase-1 Approach · discussion input, not a production spec

Prototype uses synthetic / public data only · reference architecture, to be reviewed before production